IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
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- Methodology
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Executive Summary

- This project analyzes SpaceX launch data to identify factors affecting mission success. Using SQL for data extraction, data preprocessing, feature engineering, and data visualization, we uncover key patterns and build machine learning models.
- The results highlight the most important drivers of launch outcomes and provide actionable insights for mission planning and risk management.

Introduction

SpaceX has revolutionized space exploration by developing reusable rockets and reducing launch costs. Understanding the factors that influence the success of launches is critical for mission planning, risk management, and continuous improvement of rocket technology. By analyzing historical launch data, we can identify patterns and insights that help improve future mission outcomes.

Problems We Want to Find Answers To:

- Which factors most strongly influence the success or failure of a SpaceX launch?
- Can we predict the outcome of a launch based on available data?
- How can data-driven insights help optimize mission planning and reduce risk?
- Are there trends or patterns in launch outcomes over time or by rocket type?

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
  Data was collected from CSV file available in IBM Cloud Platform
- Perform data wrangling
  The data was cleaned, encoded, and scaled, with SQL used for extraction and visualization applied to identify patterns before modeling.
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  Build classification models by selecting features and algorithms, tune hyperparameters with cross-validation, and evaluate performance using accuracy.

Data Collection

Data was downloaded from CSV file with pandas library. CSV file was available in IBM platform Cloud, initially provided by SpaceX company (shared publicly).

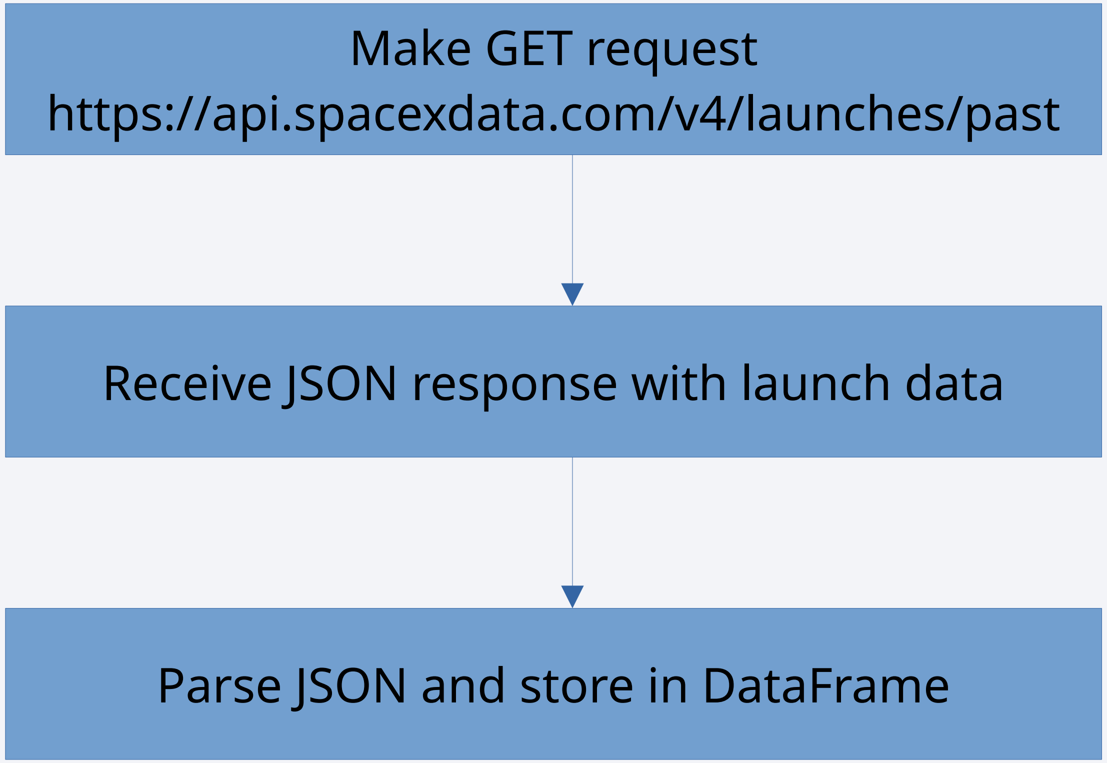
Now let's start requesting rocket launch data from SpaceX API with the following URL:

[6]
```
spacex_url="https://api.spacexdata.com/v4/launches/past"
```

Generate | + Code | + Markdown

[7]
```
response = requests.get(spacex_url)
```

Data Collection – SpaceX API



Data Collection API - link to notebook

Data Collection - Scraping

Make GET request to Wikipedia
https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922

Parse table with BeautifulSoup library

Store parsed result into Pandas DataFrame

Data Collection WebScraping - link

Data Wrangling

Filter the dataframe
include only Falcon 9 launches

↓

Cleaning and preparing data
strip strings, formatting numbers, dates

↓

Dealing with Missing Values

↓

Initial exploratory analysis

↓

Determine training labels

Data wrangling - link

EDA with Data Visualization

- Scatter plots show result of mission (success or fail) depending on:
  - flight number and payload mass
  - flight number and launch site
  - payload mass and launch site
  - flight number and orbit type
  - payload mass and orbit type
- Line plot to visualize the launch success yearly trend
- Bar plot with success rate by orbit type

EDA Data visualization - Link

EDA with SQL

- names of the unique launch sites in the space mission
- 5 records where launch sites begin with the string 'CCA'
- total payload mass carried by boosters launched by NASA (CRS)
- average payload mass carried by booster version F9 v1.1
- date when the first succesful landing outcome in ground pad was acheived
- boosters with success in drone ship with payload mass 4000-6000 kg
- total number of successful and failure mission outcomes
- booster_versions that have carried the maximum payload mass
- failure missions (month, booster version, launch site) in 2015
- ranking of missions between 2010-06-04 and 2017-03-20 by landing outcome

EDA SQL - Link

Build an Interactive Map with Folium

On the Folium map, I added markers for individual launch sites and grouped them using marker clusters to improve readability when multiple points were close together. I also created a choropleth map to visualize launch success rates by region, providing a clear geographic overview.

These objects were included to highlight both the exact locations of launches and the broader spatial patterns, making the data more insightful and easier to interpret.

Interactive Map with Folium - link

Build a Dashboard with Plotly Dash

Dashboard Summary

- Dropdown menu – allows selecting a specific launch site or all sites.
- Pie chart – shows success counts per site (or success vs. failure for a chosen site).
- Range slider – lets users filter launches by payload mass.
- Scatter plot – visualizes the correlation between payload mass and launch success, colored by booster version.

These plots and interactions were added to make the dashboard interactive and insightful, helping users explore success rates, compare sites, and understand how payload and booster type influence mission outcomes.

Interactive dashboard - Link

Predictive Analysis (Classification)

Different estimators:

- Logistic Regression
- KNN
- SVM
- Decision Tree

Prepare data
define target variable, stanardize data

Split data into train and test

Train data
find optimal parameters using GridSearchCV

Calculate accuracy and score on test data
Plot confusion matrix

Find method that performs best

Classification - Link

Results

EDA (visualization and SQL):

- success rate increased from 0 to 80% from 2013 to 2017
- most launches were to orbit GTO and ISS
- first succesful landing outcome in ground pad was in 2015
- 99/101 missions had successful outcome
- although a lot of landings failed

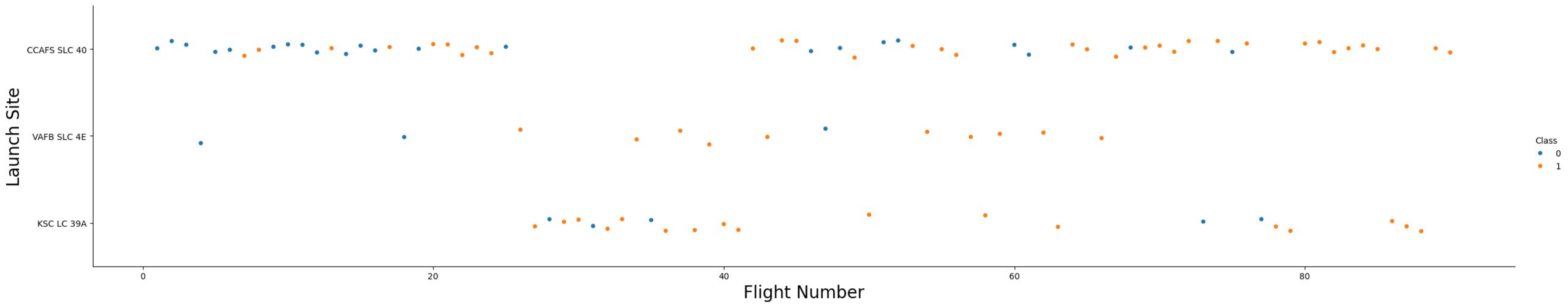
Interactive dashboards and classification:

- most success were with launch site KSC LC-39A
- no significant correlation between payload mass and success, except a lot of failures for payload mass between 500-750 kg
- best achieved score on test data was 83.3%
- all estimators achieved same score
- 100% accuracy for negative
  80% accuracy for positive cases

Section 2

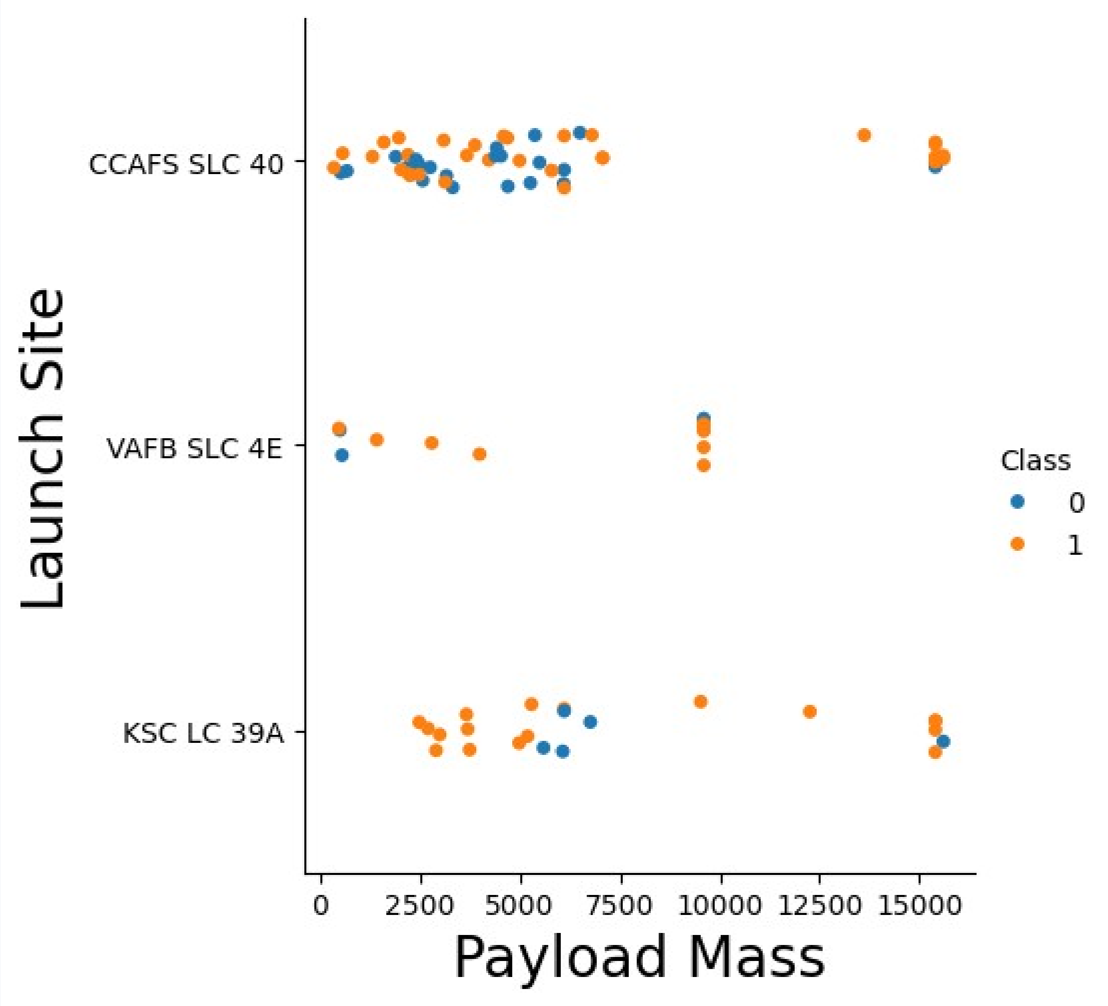
Insights drawn from EDA

Flight Number vs. Launch Site



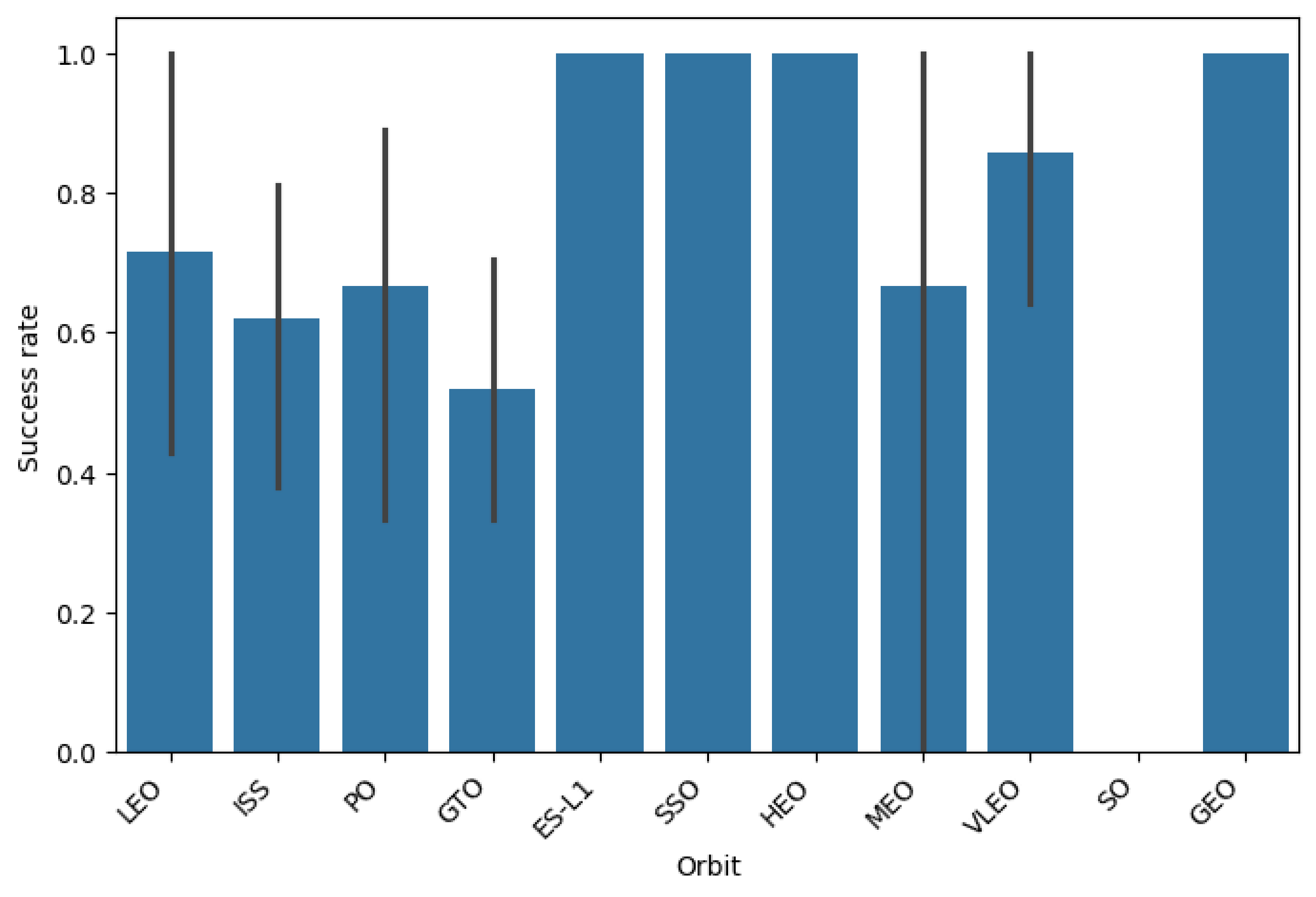
- First flights were mostly failure, launched on CCAFS SLC 40
- Recent flights were successful mostly launched on CCAFS and KSC
- VAFB was launched rarely

Payload vs. Launch Site



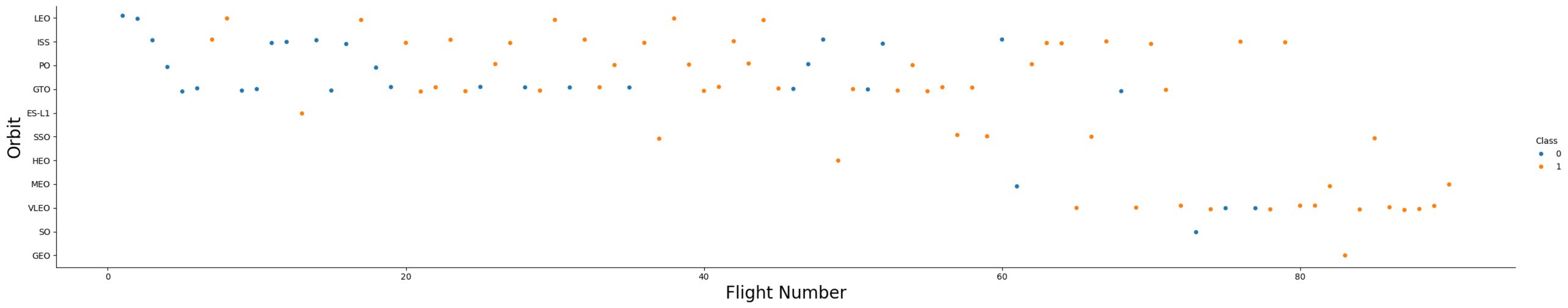
- CCAFS was mostly launched with payload mass less than 7,5t and with maximum mass
- VAFB was used for mass ~10t with success
- KSC was used similarly to CCAFS

Success Rate vs. Orbit Type



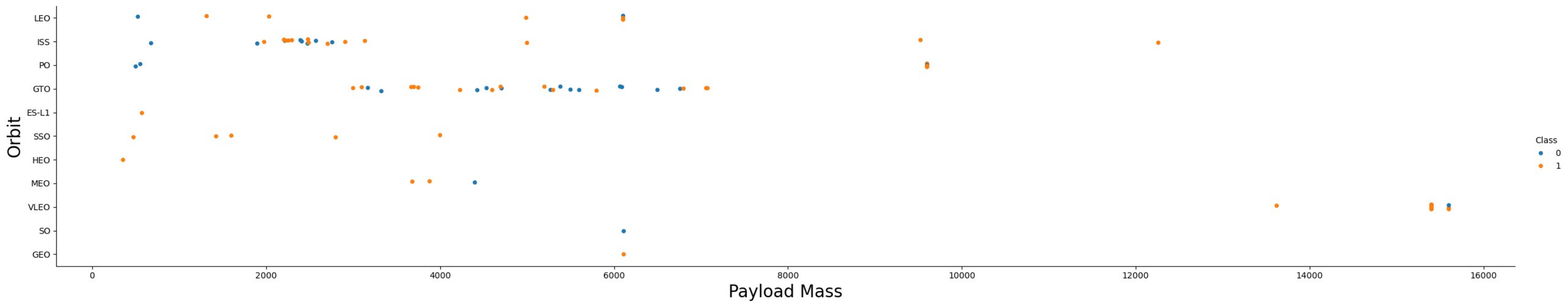
- 100% success for orbits: ES-L1, SSO, HEO, GEO
- Smallest success rate for GTO (~50%) probably because of a lot tries in early stage
- Small confidence for results for MEO orbit

Flight Number vs. Orbit Type



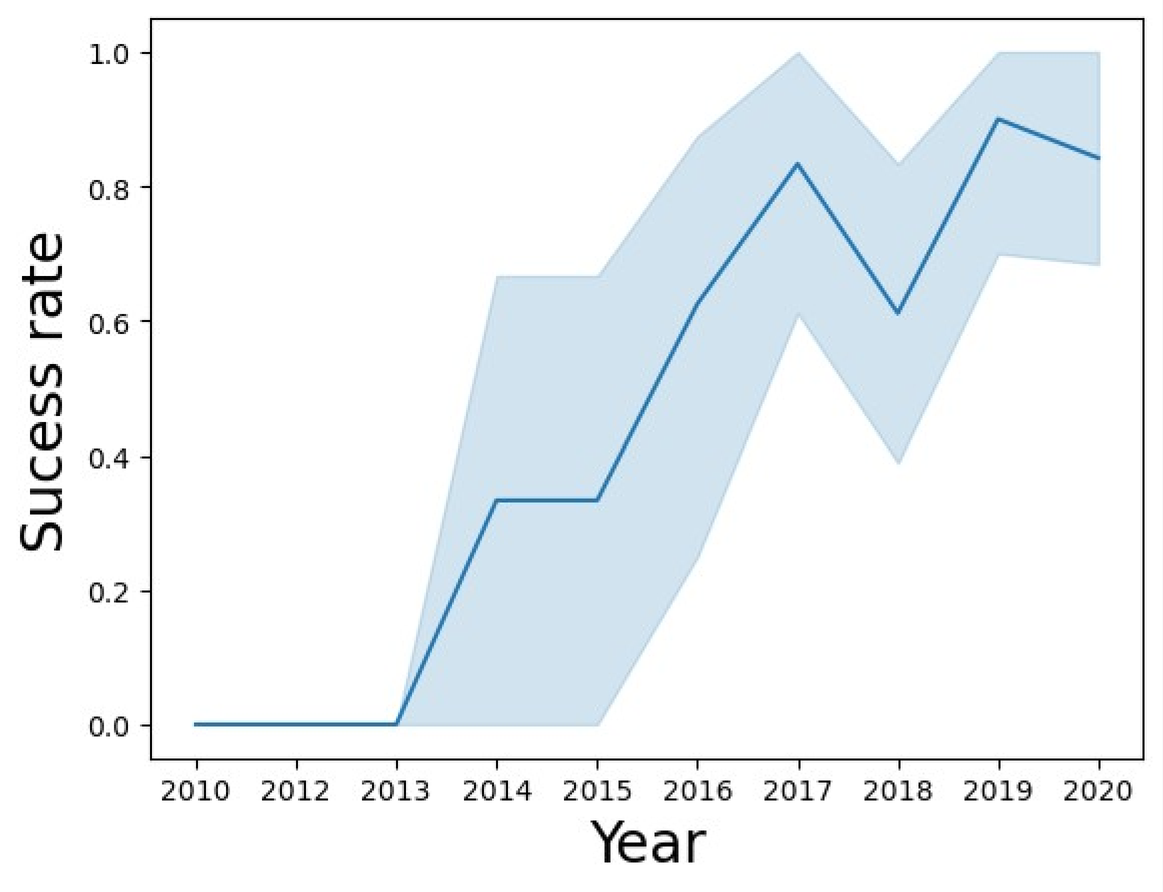
- Most recent success launches were to VLEO orbit
- First launches were mostly done on orbits: LEO, ISS, GTO, PO
- GEO orbit was launched only once with success probably due to high cost

Payload vs. Orbit Type



- Launches with high payload were only done with VLEO orbit
- Small payloads were sent to ISS orbit mostly
- Medium payloads were tested mostly on GTO orbit (with a lot of failures)

Launch Success Yearly Trend



- First success was in 2013
- Most increasing trend was between 2013 and 2017
- Average success rate in recent years was around 80% (2017-2020)

All Launch Site Names

```
select distinct(launch_site)
from spacextable
```

Zwraca unikalne nazwy
miejsc startowych
(launch_site).

Launch Site Names Begin with 'CCA'

```
Select *
from spacextable
where launch_site like 'CCA%'
limit 5
```

Gets first 5 rows from table, where launch site begins with ‘CCA’

Total Payload Mass

```
select sum(payload_mass__kg_)
from spacextable
where customer = 'NASA (CRS)'
```

Calculates total payload mass launched for customer NASA (CRS)

Average Payload Mass by F9 v1.1

```
select avg(payload_mass__kg_)
from spacextable
where booster_version = 'F9 v1.1'
```

Calculates average payload mass for booster version F9 v1.1

First Successful Ground Landing Date

```
select min(date)
from spacextable
where landing_outcome = 'Success (ground pad)'
```

Finds minimum date for landing_outcome: ‘Success (ground pad’

Successful Drone Ship Landing with Payload between 4000 and 6000

```
Select *
from spacextable
where (PAYLOAD_MASS__KG_ between 4000 and 6000)
and (Landing_Outcome = 'Success (drone ship)')
```

Selects every launch when payload mass was between 4000 and 6000 kg and landing outcome wass succesful on drone ship

Total Number of Successful and Failure Mission Outcomes

```
select mission_outcome, count(1)
from spacextable
group by mission_outcome
```

Groups launches by mission outcome, and counts frequency of groups.
Display mission_outcome and counter for every group

Boosters Carried Maximum Payload

```
select distinct(booster_version)
from spacextable
where PAYLOAD_MASS__KG_ =
      (select max(PAYLOAD_MASS__KG_) from spacextable)
```

Firstly calculates maximum payload mass, then find unique booster versions that carried this mass

2015 Launch Records

```
select substr(date,6,2) month, landing_outcome, booster_version,
launch_site
from spacextable
where substr(date,1,4) = '2015'
and landing_outcome  = 'Failure (drone ship)'
```

Selects data from year 2015 with landing outcome failure on drone ship. Then displays month, landing outcome, booster version and launch site for every launch

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
select landing_outcome, COUNT(1) AS outcome_count
from spacextable
where date BETWEEN '2010-06-04' and '2017-03-20'
group by landing_outcome
order by outcome_count DESC;
```

Ranks the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

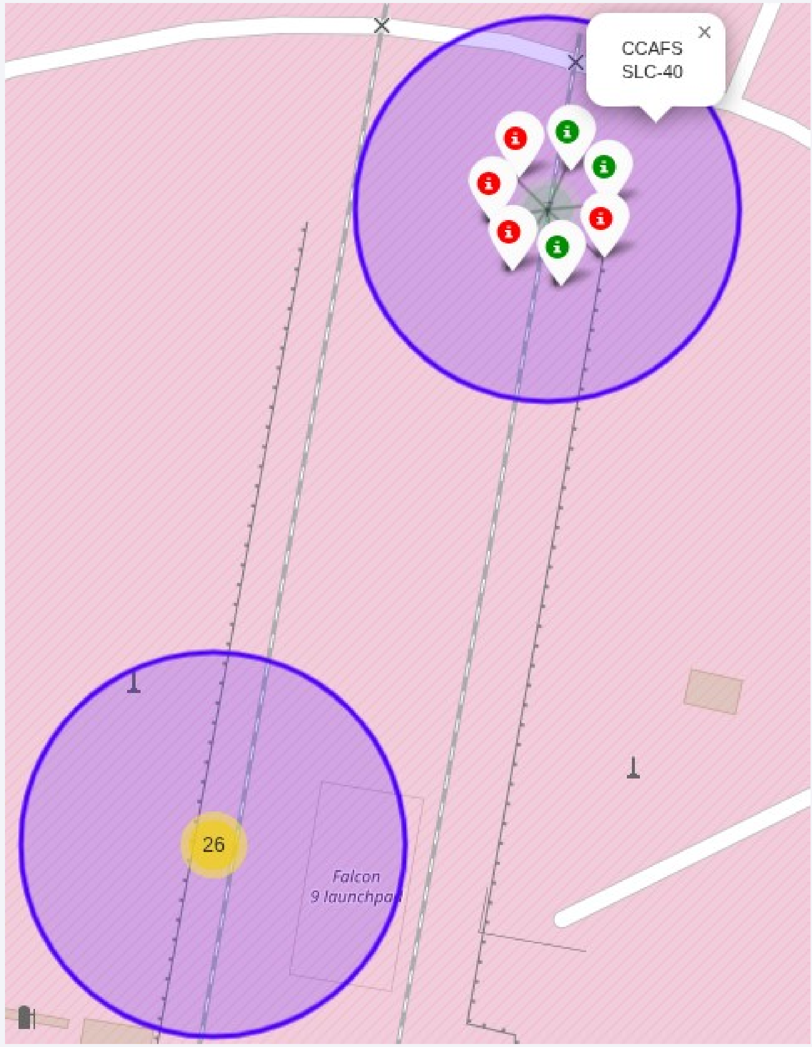
Section 3

Launch Sites Proximities Analysis

Launch Sites

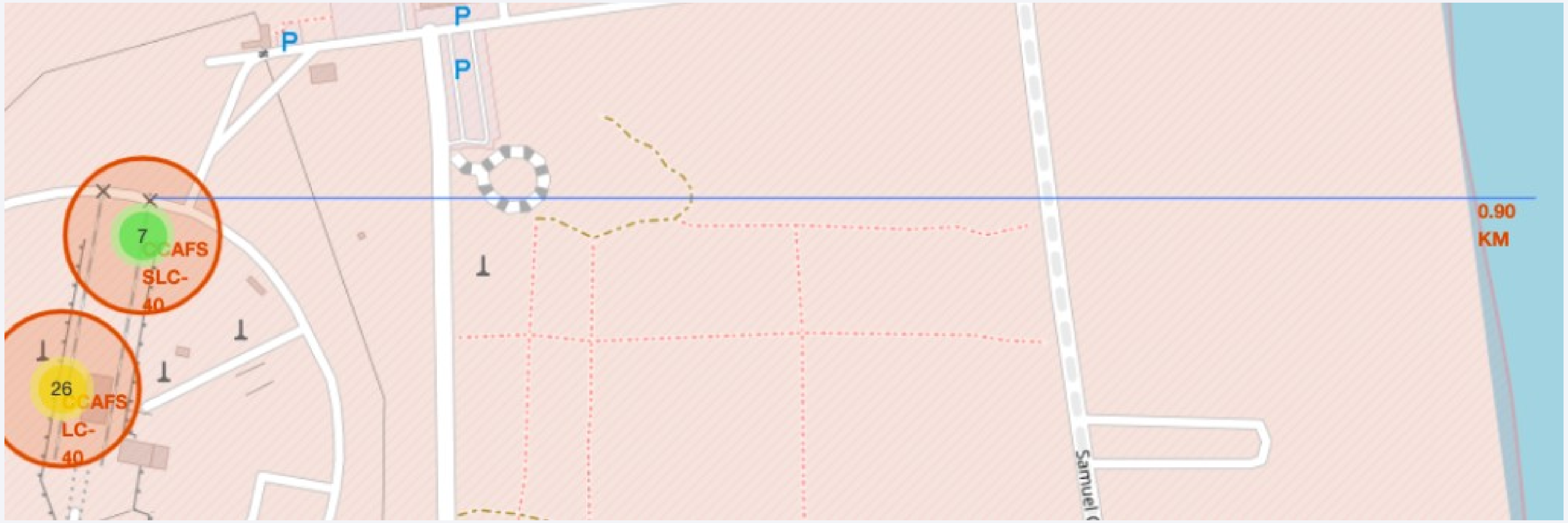
Launch Sites are located near ocean and near coasts.
Some of them are on south and some on west of USA

Success vs fail – Launch site: CCAFS SLC - 40



At CCAFS SLC-40 launch site, a little less than 50% launches were successful

Proximity to coasts, railways, highways



The selected launch site is located near a highway, a railway, and the coastline, providing good access and transport options. Distances to each feature are clearly displayed on the map, showing that the railway is slightly farther than the highway but still nearby. This proximity to multiple transport routes could be important for launch logistics and safety planning.

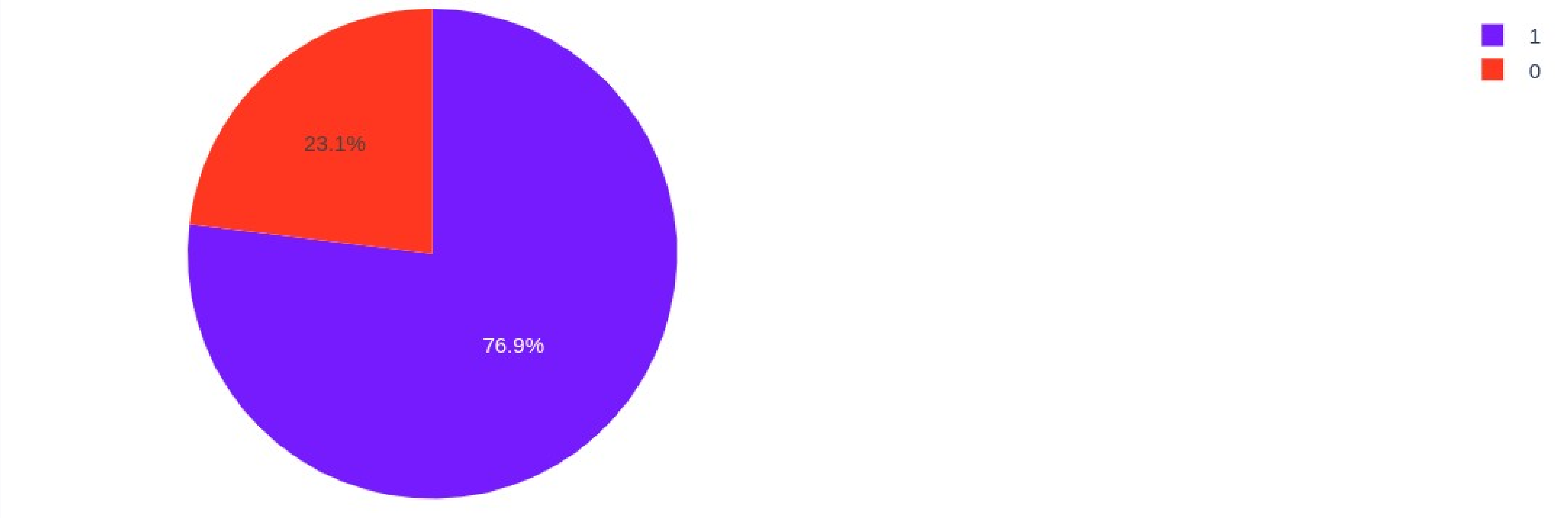
Section 4

Build a Dashboard with Plotly Dash

Success rate by launch site

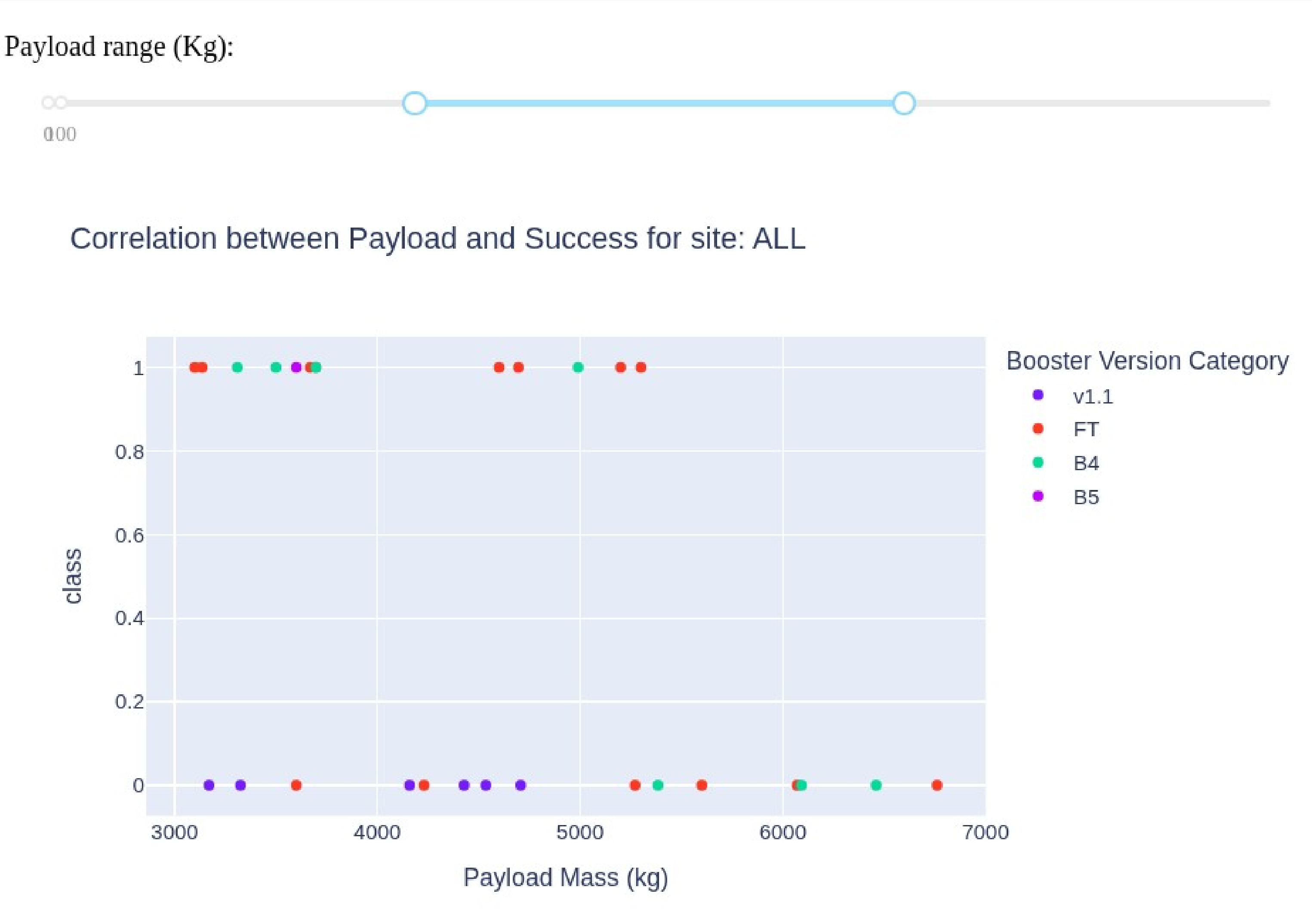
Most success launches was from launch site KSC LC-39A
29,2% of successful launches was from CCAFS LC-40
Less successes occurred for VAFB SLC-4E and CCAFS SLC-40

Total success launches for site KSC LC-39A



KSC LC-39A has success rate 76,9%.
23,1% of launches failed.

Correlation between Payload mass and success



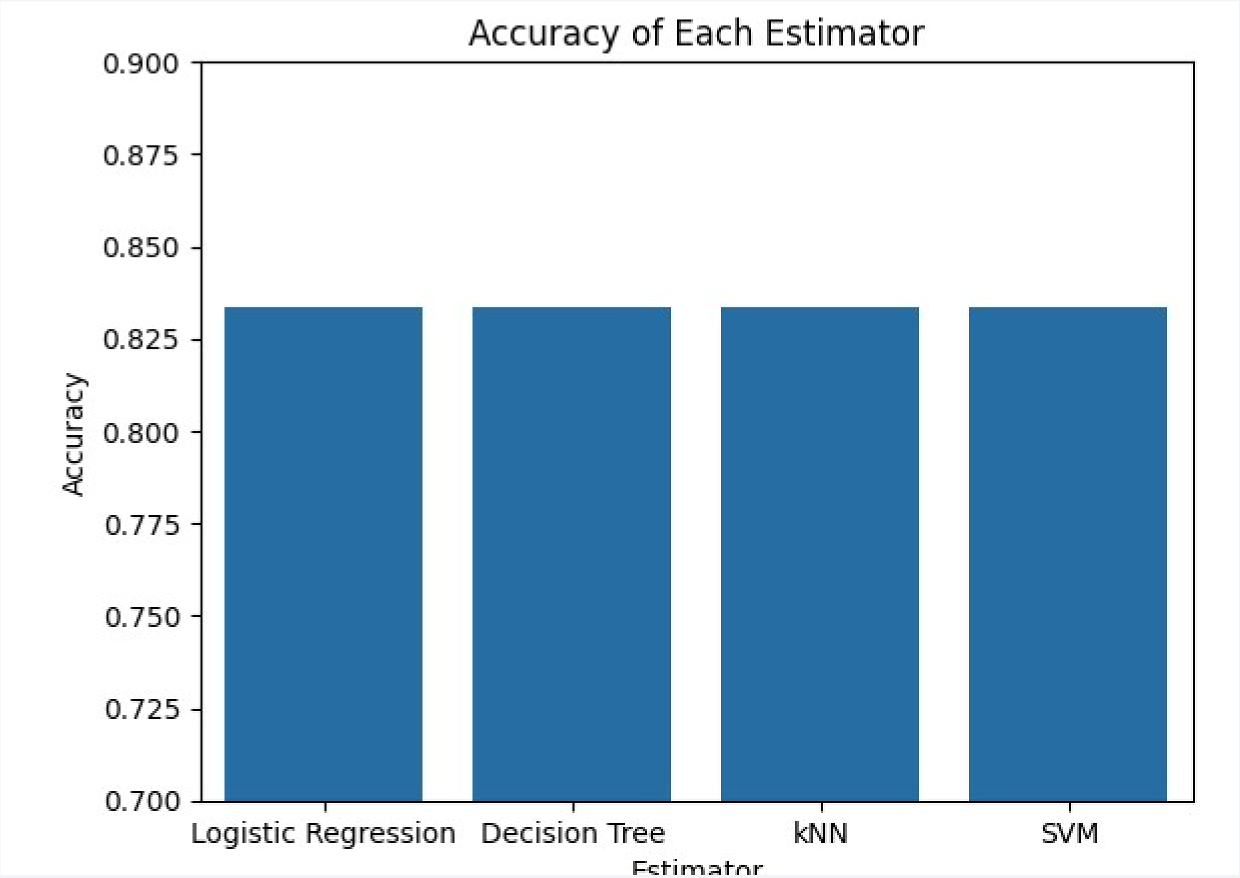
For payload mass in range 3000-7000 kg a lot of success launches were for FT Booster Ver. Category.

On the other hand a lot of fails occurred for B5

Section 5

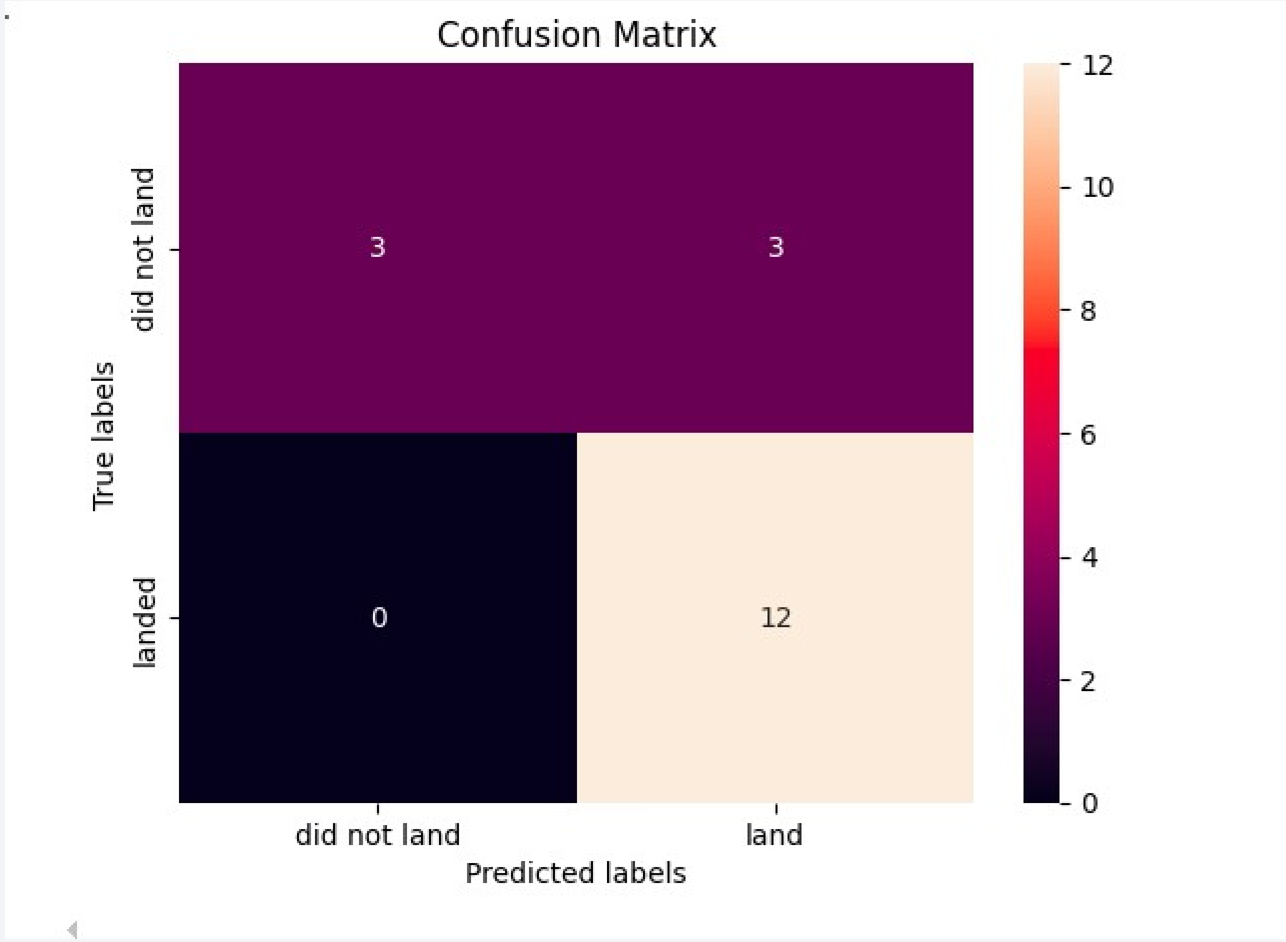
Predictive Analysis (Classification)

Classification Accuracy



All models achieved the same accuracy: 83,3%

Confusion Matrix



15/18 predictions (83,3%) were correct.

All successful landings were predicted correctly (recall is 100%)

But there is problem with negative cases. Only 50% (3 out of 6 were predicted correctly)

Conclusions

- Most SpaceX launches occurred at a few key sites, with some sites showing higher success rates than others.
- Payload mass distributions vary significantly by booster version, affecting launch outcomes.
- SQL analysis revealed that NASA (CRS) missions contributed the largest total payload mass.
- Interactive dashboards showed correlations between payload mass and success, with sliders and dropdowns enabling site-specific exploration.
- Folium maps highlighted spatial patterns: most launch sites are near highways and coastlines, while railways are slightly farther.
- Predictive models using standardized features could reasonably classify launch success, confirming the importance of payload and booster type as key predictors.

Appendix

Link to repository on github:

- https://github.com/matlewan/coursera-learning/blob/main/10-capstone-project

Thank you!